

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Sephora Beauty E-Commerce Database Management System

1. Database Design

1.1 Seller Table

Attribute	Key	Type
Seller_ID	Primary Key	-
Seller_Name	Not Null	-
Contact_Number	-	-
Email	-	-
Address	-	-

1.2 Inventory Table

Attribute	Key	Type
Inventory_ID	Primary Key	-
Product_ID	Foreign Key	-
Seller_ID	Foreign Key	-
Stock_Quantity	Default	-
Stock_Status	Default	-
Last_Updated	Not Null	-

2. Primary and Foreign Key Relationship

Seller → Inventory

- Seller.Seller_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Seller_ID → Foreign Key referencing Seller.Seller_ID
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- Relationship: One-to-Many (1:M) — one seller can have multiple inventory records.

3. Table Creation

3.1 Create Seller Table

```
CREATE TABLE Seller (  
    Seller_ID NUMBER PRIMARY KEY,  
    Seller_Name VARCHAR2(100) NOT NULL,  
    Contact_Number VARCHAR2(15),  
    Email VARCHAR2(100),
```

```
Address VARCHAR2(200)
);
```

The Seller table is created with Seller_ID as the primary key. It stores seller (beauty brand/vendor) details such as name, contact number, email, and address.

3.2 Create Inventory Table

```
CREATE TABLE Inventory (
    Inventory_ID NUMBER PRIMARY KEY,
    Seller_ID NUMBER NOT NULL,
    Product_ID NUMBER NOT NULL,
    Stock_Quantity NUMBER DEFAULT 0,
    Stock_Status VARCHAR2(20) DEFAULT 'In Stock',
    Last_Updated DATE NOT NULL,
    CONSTRAINT fk_seller FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID),
    CONSTRAINT fk_product FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
);
```

The Inventory table is created with Inventory_ID as the primary key. Seller_ID and Product_ID are foreign keys that establish relationships with the Seller and Product tables.

4. Data Insertion

4.1 Seller Records

Five seller records (beauty brand sellers) are inserted into the Seller table, followed by a COMMIT to save the transaction.

```
INSERT INTO Seller VALUES (1, 'Glow & Co Beauty', '9998887771', 'glowco@gmail.com',
    'Chennai');
1 row created.

INSERT INTO Seller VALUES (2, 'Luxe Cosmetics', '9998887772',
    'luxecosmetics@gmail.com', 'Bangalore');
1 row created.

INSERT INTO Seller VALUES (3, 'Pure Skin Essentials', '9998887773',
    'pureskin@gmail.com', 'Mumbai');
1 row created.

INSERT INTO Seller VALUES (4, 'Radiant Beauty Supply', '9998887774',
    'radiantbeauty@gmail.com', 'Delhi');
1 row created.
```

```
INSERT INTO Seller VALUES (5, 'Velvet Glow Beauty', '9998887775',  
'velvetglow@gmail.com', 'Hyderabad');  
1 row created.
```

```
COMMIT;
```

4.2 Inventory Records

Inventory records are inserted with seller IDs, product IDs, stock quantities, stock status, and last updated dates.

```
INSERT INTO Inventory VALUES (301, 1, 201, 40, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (302, 1, 202, 0, 'Out of Stock', TO_DATE('15-09-  
2026','DD-MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (303, 2, 203, 25, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (304, 2, 204, 10, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (305, 3, 205, 0, 'Out of Stock', TO_DATE('15-09-  
2026','DD-MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (306, 3, 206, 18, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (307, 4, 207, 12, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (308, 4, 208, 0, 'Out of Stock', TO_DATE('15-09-  
2026','DD-MM-YYYY'));  
1 row created.
```

```
INSERT INTO Inventory VALUES (309, 5, 209, 30, 'In Stock', TO_DATE('15-09-2026','DD-  
MM-YYYY'));  
1 row created.
```

Attribute	Key	Type
201	Matte Foundation	Makeup
202	Velvet Lipstick	Makeup
203	Volumizing Mascara	Makeup
204	Hydrating Moisturizer	Skincare
205	Vitamin C Serum	Skincare

```
INSERT INTO Inventory VALUES (310, 5, 210, 0, 'Out of Stock', TO_DATE('15-09-2026', 'DD-MM-YYYY'));
1 row created.

COMMIT;
```

Note: this assumes a Product table (Product_ID, Product_Name, Category, Price) already exists in the Sephora database, following the same beauty product catalog used in the earlier ER diagram and Customer-table work. Adjust the Product_IDs below if your existing Product table uses different values.

5. Data Retrieval

5.1 Display Seller Records

All seller records are retrieved to verify successful insertion.

```
SELECT * FROM Seller;
```

SELLER_ID	SELLER_NAME	CONTACT_NUMBER	EMAIL	ADDRESS
-----	-----	-----	-----	-----
-				
1	Glow & Co Beauty	9998887771	glowco@gmail.com	Chennai
2	Luxe Cosmetics	9998887772	luxecosmetics@gmail.com	Bangalore
3	Pure Skin Essentials	9998887773	pureskin@gmail.com	Mumbai
4	Radiant Beauty Supply	9998887774	radiantbeauty@gmail.com	Delhi
5	Velvet Glow Beauty	9998887775	velvetglow@gmail.com	Hyderabad

```
5 rows selected.
```

5.2 Display Inventory Records

All inventory records are retrieved to verify the stock information.

```
SELECT * FROM Inventory;
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
-----	-----	-----	-----	-----	-----
301	1	201	40	In Stock	15-SEP-26
302	1	202	0	Out of Stock	15-SEP-26

303	2	203	25	In Stock	15-SEP-26
304	2	204	10	In Stock	15-SEP-26
305	3	205	0	Out of Stock	15-SEP-26
306	3	206	18	In Stock	15-SEP-26
307	4	207	12	In Stock	15-SEP-26
308	4	208	0	Out of Stock	15-SEP-26
309	5	209	30	In Stock	15-SEP-26
310	5	210	0	Out of Stock	15-SEP-26

10 rows selected.

6. Seller Product Information

A JOIN query is used to display the seller name, product name, stock quantity, and stock status.

```
SELECT
    s.Seller_Name,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Seller s
JOIN Inventory i ON s.Seller_ID = i.Seller_ID
JOIN Product p ON i.Product_ID = p.Product_ID
ORDER BY s.Seller_Name;
```

SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
-----	-----	-----	-----
Glow & Co Beauty	Matte Foundation	40	In Stock
Glow & Co Beauty	Velvet Lipstick	0	Out of Stock
Luxe Cosmetics	Volumizing Mascara	25	In Stock
Luxe Cosmetics	Hydrating Moisturizer	10	In Stock
Pure Skin Essentials	Vitamin C Serum	0	Out of Stock
Pure Skin Essentials	Eyeshadow Palette	18	In Stock
Radiant Beauty Supply	Floral Perfume	12	In Stock
Radiant Beauty Supply	Glossy Nail Polish	0	Out of Stock
Velvet Glow Beauty	Sunscreen SPF50	30	In Stock
Velvet Glow Beauty	Hair Repair Serum	0	Out of Stock

10 rows selected.

7. Product Availability

7.1 Available Products

Products with stock available are displayed using the Stock_Status field.

```
SELECT
```

```

    p.Product_ID,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Product p
JOIN Inventory i ON p.Product_ID = i.Product_ID
WHERE i.Stock_Status = 'In Stock';

```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
201	Matte Foundation	40	In Stock
203	Volumizing Mascara	25	In Stock
204	Hydrating Moisturizer	10	In Stock
206	Eyeshadow Palette	18	In Stock
207	Floral Perfume	12	In Stock
209	Sunscreen SPF50	30	In Stock

6 rows selected.

7.2 Unavailable Products

Products with zero stock or Out of Stock status are displayed separately.

```

SELECT
    p.Product_ID,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Product p
JOIN Inventory i ON p.Product_ID = i.Product_ID
WHERE i.Stock_Status = 'Out of Stock';

```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
202	Velvet Lipstick	0	Out of Stock
205	Vitamin C Serum	0	Out of Stock
208	Glossy Nail Polish	0	Out of Stock
210	Hair Repair Serum	0	Out of Stock

4 rows selected.

8. Inventory Update

Demonstrate updating stock information for products that were restocked.

```

UPDATE Inventory

```

```

SET Stock_Quantity = 20,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 305;
1 row updated.

UPDATE Inventory
SET Stock_Quantity = 15,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 310;
1 row updated.

SELECT * FROM Inventory WHERE Inventory_ID IN (305, 310);

```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
-----	-----	-----	-----	-----	-----
305	3	205	20	In Stock	17-SEP-26
310	5	210	15	In Stock	17-SEP-26

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SELECT
    i.Inventory_ID,
    s.Seller_Name,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status,
    i.Last_Updated
FROM Inventory i
JOIN Seller s ON i.Seller_ID = s.Seller_ID
JOIN Product p ON i.Product_ID = p.Product_ID
ORDER BY i.Inventory_ID;

```

(Output follows the same layout as Section 5.2, updated with the Section 8 changes to Inventory_ID 305 and 310.)

9.2 Total Stock by Seller

```

SELECT
    s.Seller_Name,
    SUM(i.Stock_Quantity) AS Total_Stock
FROM Seller s
JOIN Inventory i ON s.Seller_ID = i.Seller_ID

```

```
GROUP BY s.Seller_Name
ORDER BY s.Seller_Name;
```

SELLER_NAME	TOTAL_STOCK
-----	-----
Glow & Co Beauty	40
Luxe Cosmetics	35
Pure Skin Essentials	38
Radiant Beauty Supply	12
Velvet Glow Beauty	45

9.3 Products with Zero Stock

```
SELECT
    p.Product_ID,
    p.Product_Name,
    s.Seller_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Product p
JOIN Inventory i ON p.Product_ID = i.Product_ID
JOIN Seller s ON i.Seller_ID = s.Seller_ID
WHERE i.Stock_Quantity = 0;
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
-----	-----	-----	-----	-----
202	Velvet Lipstick	Glow & Co Beauty	0	Out of Stock
208	Glossy Nail Polish	Radiant Beauty Supply	0	Out of Stock

2 rows selected.
(after the Section 8 updates – 205 and 210 no longer appear here)

9.4 Available and Unavailable Product Count

```
SELECT Stock_Status, COUNT(*) AS Total_Product
FROM Inventory
GROUP BY Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCT
-----	-----
In Stock	8
Out of Stock	2

10. Conclusion

The Seller and Inventory tables were successfully created and connected with the existing Sephora Product table using primary and foreign keys. Seller (vendor) and product information can be maintained along with stock quantities and stock status. Available and unavailable products can be tracked using SQL queries, and inventory can be kept current through UPDATE statements. Inventory reports such as the complete inventory listing, seller-wise total stock, zero-stock products, and availability counts can also be generated to support restocking decisions.